

Problem 1

$$|M| = 46$$

$$|T| = 35$$

$$|M \cap T| = 18$$

(a) We can write, $M = (M - T) \cup (M \cap T)$

Now, let an arbitrary x st,

$$x \in M - T$$

$$\Rightarrow x \in M \wedge x \notin T$$

$$\Rightarrow x \notin T$$

$$\Rightarrow x \notin M \cap T$$

$\therefore M - T$ and $M \cap T$ are disjoint.

$$\therefore |M| = |M - T| + |M \cap T|$$

$$|M - T| = |M| - |M \cap T| = 46 - 18 = 28$$

By similar argument, $T - M$ and $M \cap T$ are disjoint.

$$\therefore |T| = |T - M| + |M \cap T|$$

$$|T - M| = 35 - 18 = 17$$

$$\text{Again, } M \cup T = (M - T) \cup (M \cap T) \cup (T - M)$$

$$|M \cup T| = |M - T| + |M \cap T| + |T - M|$$

$$= 28 + 17 + 18$$

$$= 63$$

[all 3 are mutually disjoint]

(ib)

By definition, $x \in E \Rightarrow (x \in M \wedge x \notin T) \vee (x \notin M \wedge x \in T)$
for an arbitrary x .

$$\therefore E = (M-T) \cup (T-M)$$

$$\begin{aligned}\therefore |E| &= |M-T| \cup |T-M| \\ &= 28 + 17 \\ &= 45\end{aligned}$$

[proof of disjointness
in ques later]

PS: Proof that $M-T$ and $T-M$ are disjoint:

Let, an arbitrary x st,

$$\begin{aligned}x &\in (M-T) \cap (T-M) \\ \Rightarrow (x \in M \wedge x \notin T) \wedge (x \in T \wedge x \notin M)\end{aligned}$$

But, $x \in M \wedge x \notin M \Rightarrow$ no such x exists.

$$\therefore (M-T) \cap (T-M) = \phi$$

$\therefore M-T$ and $T-M$ are disjoint.

$$(c) \quad \text{Given, } |C| = 20 \quad C \subseteq E \\ |C \cap M| = 9$$

Since, E can be written as a ^{disjoint} union, C can also be written as a disjoint union.

Since, $x \in E \Rightarrow x \in M$ or $x \in T$ but not both, C (and also E) are partitioned by M and T .

Formally, $(x \in C) \wedge (x \notin C \cap M) \Rightarrow x \in T$. But, we also know, $(x \notin C \cap M) \Rightarrow$ ~~not in~~ $x \in C-M$.

$$\text{As before, } C = (C-M) \cup (C \cap M)$$

$$|C| = |C-M| + |C \cap M|$$

$$20 = |C-M| + 9 \Rightarrow |C-M| = 11$$

By partitioning argument,

$$C = (C \cap M) \cup (C \cap T)$$

$$|C| = |C \cap M| + |C \cap T|$$

$$|C \cap T| = 20 - 9 = 11$$

(d) By definition, $P(E)$ = set of all subsets of E .

$\therefore C \subseteq E \Rightarrow C \in P(E)$ is true trivially.

Problem 2:

(a) Proving $A - (B \cup C) \subseteq (A - B) \cap (A - C)$:

Let an arbitrary $x \in A - (B \cup C)$

$$\Rightarrow (x \in A) \wedge (x \notin B \cup C)$$

$$\Rightarrow (x \in A) \wedge (x \notin B \wedge x \notin C)$$

$$\Rightarrow (x \in A \wedge x \notin B) \wedge (x \in A \wedge x \notin C)$$

$$\Rightarrow (x \in A - B) \wedge (x \in A - C)$$

$$\Rightarrow x \in (A - B) \cap (A - C)$$

$$\therefore A - (B \cup C) \subseteq (A - B) \cap (A - C)$$

Proving the other direction:

$$x \in (A - B) \cap (A - C)$$

$$\Rightarrow (x \in A - B) \wedge (x \in A - C)$$

$$\Rightarrow (x \in A \wedge x \notin B) \wedge (x \in A \wedge x \notin C)$$

$$\Rightarrow (x \in A) \wedge (x \notin B) \wedge (x \notin C)$$

$$\Rightarrow (x \in A) \wedge \neg(x \in B \vee x \in C)$$

$$\Rightarrow (x \in A) \wedge \neg(x \in B \cup C)$$

$$\Rightarrow (x \in A) \wedge (x \notin B \cup C)$$

$$\Rightarrow x \in A - (B \cup C)$$

$$\therefore (A - B) \cap (A - C) \subseteq A - (B \cup C)$$

$$\text{Therefore, } A - (B \cup C) = (A - B) \cap (A - C)$$

(16) Proving $A \subseteq B \Rightarrow A \cup B = B$:

$$A \subseteq B \Rightarrow (x \in A \rightarrow x \in B)$$

[by definition]

Let, an arbitrary $x \in A \cup B$

$$\Rightarrow x \in A \vee x \in B$$

$$\Rightarrow x \in B \vee x \in B$$

$$\Rightarrow x \in B$$

$$\therefore A \cup B \subseteq B$$

Let, an arbitrary $x \in B$

$$\Rightarrow x \in A \cup B$$

[by definition]

$$\therefore B \subseteq A \cup B$$

$$\therefore A \subseteq B \Rightarrow A \cup B = B$$

Proving $A \cup B = B \Rightarrow A \subseteq B$,

Let, an arbitrary $x \in A$

$$\Rightarrow x \in A \cup B$$

$$\Rightarrow x \in B$$

[definition]

$$\therefore A \subseteq B$$

[definition]

$$\therefore A \cup B = B \Rightarrow A \subseteq B$$

$$\therefore A \subseteq B \iff A \cup B = B.$$

Problem 3:

(a) Given, $X = \{a, b, c\}$

$P(X) :=$ set of all subsets of X

$$= \{ \{ \}, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\} \}$$

To get $|P(P(X))|$ we need to prove part (b).

(b) $P(S) :=$ set of all subsets of S

While constructing $A \subseteq S$, there are 2 choices of each element $x \in S$,

(1) in: $x \in A$ or

(2) out: $x \notin A$

Since, there are n elements, number of possible subsets = $\underbrace{2 \times 2 \times \dots \times 2}_{n\text{-times}} = 2^n$

$$\therefore |P(S)| = 2^{|S|} = 2^n$$

(continuing (a)) $|P(P(X))| = 2^{|P(X)|} = 2^8 = 256$

(c) In cumulative hierarchy, every set that belongs to stage V_α , is made of elements belonging to an earlier stage V_β set, $\beta < \alpha$.

Therefore, if $x \in x$, x would need to occur at an earlier stage than itself, which is impossible.

$\therefore$ a set can't contain itself.

Problem 4:

(a) Talking about $f_1: \mathbb{Z} \rightarrow \mathbb{Z}$, $f_1(x) = x^2$,

f_1 isn't surjective because only perfect squares are mapped AND no negative integer is mapped.

f_1 isn't injective because $f(x) = f(-x)$ but $x \neq -x$ ($\forall x \neq 0$).

$\therefore f_1$ is neither surjective nor injective.

Talking about $f_2: \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$, $f_2(x) = x^2$

f_2 is injective. Proof:

$$\begin{aligned} \text{Suppose } f_2(x_1) &= f_2(x_2) \\ \Rightarrow x_1^2 &= x_2^2 \end{aligned}$$

Since, $x_1, x_2 \geq 0$, we can take square root

$$\Rightarrow x_1 = x_2$$

$$\therefore f_2(x_1) = f_2(x_2) \Rightarrow x_1 = x_2$$

f_2 is surjective. Proof -

To prove: For an $y \in \mathbb{R}_{\geq 0}$, we need an $x \in \mathbb{R}_{\geq 0}$ st $f_2(x) = y$.

We can choose $x = \sqrt{y}$ because $y \geq 0$.

$$\text{Then, } f_2(x) = (\sqrt{y})^2 = y$$

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$\therefore$ Every $y \in \mathbb{R}_{>0}$ has a preimage
 $\therefore f_2$ is surjective.

Therefore, f_2 is bijective.

(b) We can take $f_1: \mathbb{N} \rightarrow \mathbb{N}$, $f_1(x) = 2x$.

f_1 is injective because $f_1(a) = f_1(b) \Rightarrow 2a = 2b \Rightarrow a = b$.
 But, f_1 is not surjective because odd numbers are not mapped. Ex-

$\nexists x \in \mathbb{N}$ such that $f_1(x) = 2x = 3$.

We can take $f_2: \mathbb{N} \rightarrow \mathbb{N}$, $f_2(x) = \begin{cases} 1, & x=1 \\ x-1, & x \geq 2 \end{cases}$

It is not injective because $f_2(1) = f_2(2) = 1$ but $1 \neq 2$.

It is surjective because all natural numbers are mapped as:

$$\boxed{\cancel{n+1} \mapsto n \mapsto \cancel{n-1} \mapsto n \mapsto \cancel{n} \mapsto n-1 \quad \forall n \geq 2}$$

If $y=1$, choose $x=1$ so that $f(1)=1$.

If $y \geq 2$, choose $x=y+1$ so that $f(y+1) = (y+1)-1 = y$.

(c) Given, $h: \mathbb{R} \rightarrow \mathbb{R}$ $h(x) = 2x+3$

Proof of injectivity:

$$h(x_1) = h(x_2)$$

$$2x_1 + 3 = 2x_2 + 3$$

$$2x_1 = 2x_2$$

$$x_1 = x_2$$

$$\therefore h(x_1) = h(x_2) \Rightarrow x_1 = x_2$$

Proof of surjectivity:

Let, $y \in \mathbb{R}$

We need $x \in \mathbb{R}$ st $h(x) = y$

$$2x + 3 = y$$

$$2x = y - 3$$

$$x = \frac{y-3}{2}$$

Since $y \in \mathbb{R}$, $\frac{y-3}{2} \in \mathbb{R}$.

Clearly, $h(x) = y$ when $x = \frac{y-3}{2}$.

$\therefore \exists x \in \mathbb{R}$ st $h(x) = y$, $\forall y \in \mathbb{R}$.

$\therefore h(x)$ is surjective.

$\therefore h(x)$ is bijective.

Problem 6:

Given:

$$A = \mathbb{R} \setminus \{1\}$$

$$f: A \rightarrow A, \quad f(x) = \frac{x+1}{x-1}$$

$$g: \mathbb{R} \rightarrow \mathbb{R}, \quad g(x) = x^2$$

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$$\begin{aligned}
 (a) \quad (f \circ f)(x) &= f(f(x)) \\
 &= f\left(\frac{x+1}{x-1}\right) \\
 &= \frac{\frac{x+1}{x-1} + 1}{\frac{x+1}{x-1} - 1} \\
 &= \frac{(x+1) + (x-1)}{(x+1) - (x-1)} \\
 &= \frac{2x}{2} \\
 &= x
 \end{aligned}$$

(b) We know, if $f: A \rightarrow A$, $f(a)=b$, $a, b \in A$
 then $f^{-1}: A \rightarrow A$, $f^{-1}(b)=a$

$$\begin{aligned}
 \therefore f(f^{-1}(b)) &= b \\
 f(f^{-1}(x)) &= x
 \end{aligned}$$

[replacing the symbol for better notation]

$$f(f(f^{-1}(x))) = f(x)$$

$$f^{-1}(x) = f(x)$$

[we know $(f \circ f)(x) = x$]

(c)

$$\begin{aligned}
 (g \circ f)(x) &= g(f(x)) \\
 &= g\left(\frac{x+1}{x-1}\right) \\
 &= \left(\frac{x+1}{x-1}\right)^2
 \end{aligned}$$

$\forall x \in A$

Check for injectivity:

Suppose $(g \circ f)(x_1) = (g \circ f)(x_2)$

$$\left(\frac{x_1+1}{x_1-1} \right)^2 = \left(\frac{x_2+1}{x_2-1} \right)^2$$

$$\frac{x_1+1}{x_1-1} = \pm \left(\frac{x_2+1}{x_2-1} \right)$$

$$\text{If } \frac{x_1+1}{x_1-1} = -\frac{x_2+1}{x_2-1},$$

$$(x_1+1)(x_2-1) = -(x_2+1)(x_1-1)$$

$$x_1x_2 + x_2 - x_1 - 1 = -x_2x_2 + x_2 - x_2 + 1$$

$$2x_1x_2 = 2$$

$$x_1x_2 = 1$$

$$x_1 = \frac{1}{x_2}$$

Since $x_1, x_2 \neq 1$, $x_1 \neq x_2$.

~~Therefore~~

$\therefore \exists x_1, x_2 \in A$ such that, $(g \circ f)(x_1) = (g \circ f)(x_2)$ but $x_1 \neq x_2$; specifically when $x_1x_2 = 1$.

$\therefore g \circ f$ is not injective.

Problem 5:

(a) ~~We~~ We define $f: \mathbb{Z} \rightarrow \mathbb{E}$,

$$f(x) := 4x$$

Injectivity:

$$\begin{aligned} \text{Suppose } f(x_1) &= f(x_2) \\ 4x_1 &= 4x_2 \\ x_1 &= x_2 \end{aligned}$$

$$\therefore f(x_1) = f(x_2) \Rightarrow x_1 = x_2$$

Surjectivity:

Let, a $y \in \mathbb{E}$.

Then, $y = 4k$ for some $k \in \mathbb{Z}$.

We need $x \in \mathbb{Z}$ st, $f(x) = y$

We can take $x = k$.

$$\text{Then, } f(k) = 4k = y$$

$\therefore \exists x \in \mathbb{Z}$ such that $f(x) = y, \forall y \in \mathbb{E}$.

$\therefore f$ is ~~surjective~~ bijective. $\therefore |\mathbb{Z}| = |\mathbb{E}| \therefore \mathbb{E}$ is countably infinite.

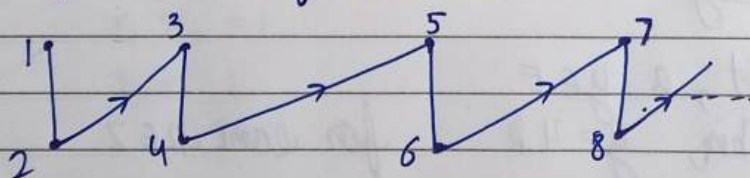
(b) Set $\mathbb{E} \times \mathbb{E}_{0,1}$ is of the form:

$$\{(0,0), (0,1), (4,0), (4,1), (-4,0), (-4,1), (8,0), (8,1), \dots\}$$

We can arrange the elements into 2 rows:

$$\begin{array}{cccc} (0,0) & (4,0) & (-4,0) & (8,0) \dots \\ (0,1) & (4,1) & (-4,1) & (8,1) \dots \end{array}$$

And map them as follows to natural numbers:



It is clear that no tuple repeats and, each natural is mapped.

Since $\mathbb{E} \times \mathbb{E}_{0,1}$ can be mapped to $\mathbb{N}$, it is countably infinite.